

Optenni 3-port isolation product baseline

Native PDF traceability report - PDF schema 1 - HTML schema 1 - project updated 2026-07-14T15:05:23.970035+00:00

Input	3_antennas.s3p	Ports / points	3 / 1001
Frequency range	1000.000 - 3000.000 MHz	Reference resistance	50.0 ohm
Software	RF Matching Engine API 2.0.0 / core 0.2.0	Project schema	v2 (migrated from v1)
Touchstone SHA-256	4af62325474067d752dac32138c447b1da5e714 a4140bafed484f0649131f6bc	Project integrity	05c763d4698249a767fd7cf3bc08def0d351d38 42e834699671e2316b1514be4

Candidate comparison

Candidate	Score	Average eta	Minimum eta	Component loss	BOM	Yield	Confidence interval
1 selected	55.2%	54.6%	40.5%	15.14%	3	-	-
2	54.7%	53.7%	40.3%	15.64%	3	-	-
3	54.6%	53.2%	40.4%	15.38%	3	-	-
4	54.5%	54.6%	40.5%	15.08%	4	-	-
5	54.5%	54.5%	40.5%	15.16%	4	-	-
6	54.4%	54.3%	40.4%	15.28%	4	-	-
7	54.9%	52.3%	43.7%	22.35%	3	-	-
8	55.1%	52.7%	43.9%	22.61%	3	-	-
9	55.1%	53.4%	45.0%	22.13%	4	-	-
10	55.2%	53.6%	45.2%	22.05%	4	-	-

Selected solution

Mode / objective	joint / balanced	Average total efficiency	54.6%
Minimum total efficiency	40.5%	Component loss	15.14%
Power balance error	-	Efficiency basis	total_efficiency_estimate
Manufacturing yield	-	Yield confidence interval	-
P5 score margin	-	Loss-aware ideal seed	-
Ideal seed values	-	Ideal seed score / evaluations	- / -
Measured search path	not applicable	Physical evaluations / models	- / -
Active physical points	-	Maximum components searched	-

Bill of materials and topology

Port	Position	Connection	Type	Part number	Value
Port 1	1	shunt	capacitor	GCM0334C1ER50WA01	0.5pF
Port 2	1	shunt	capacitor	GQM2195C2E131GB12	130.0pF
Port 3	1	shunt	inductor	LQW18CA32NJ00	32.0nH

Configuration and state assignment

Configuration	Active bands	State	Score
Not a state-configured solution	-	-	-

Configuration yield

Configuration	State	Individual yield
Single configuration or no saved state yield	-	-

Isolation targets

Path	Band	Limit	Worst	Average	Status
S21	-	-20.00 dB	-23.68 dB	-28.46 dB	PASS

Stored full-physical curves

Port 1 return loss



Port 1 total efficiency



Port 2 return loss



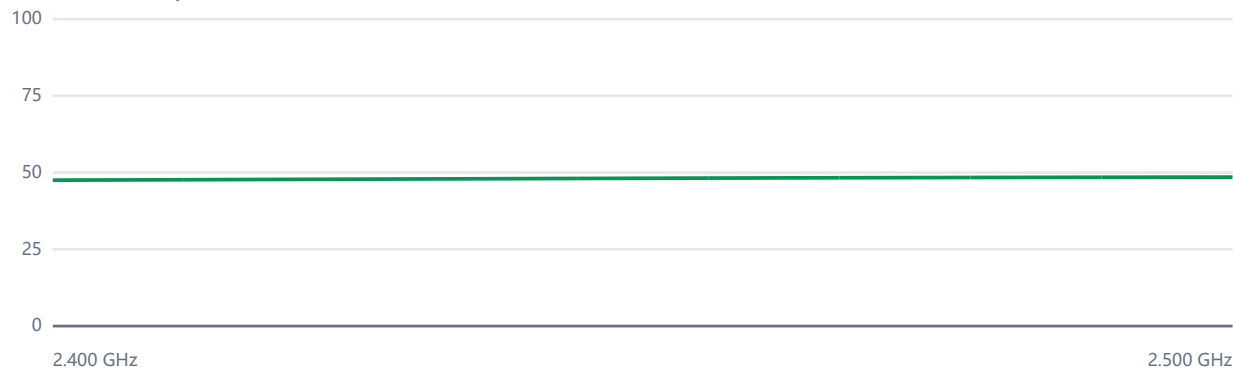
Port 2 total efficiency



Port 3 return loss



Port 3 total efficiency



Optimization configuration

```
{
  "band_state_map": null,
  "beam_width": 10,
  "component_series": null,
  "debug": false,
  "debug_top_n": 10,
  "isolation_targets": [
    {
      "average_weight": 0.0,
      "band_mhz": [
        2400.0,
        2500.0
      ],
      "destination_port": 1,
      "maximum_db": -20.0,
      "source_port": 0,
      "weight": 1.0
    }
  ],
  "mode": "joint",
  "num_band_points": 10,
  "objective": "balanced",
  "ports": [
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      "bands_mhz": [
        [
          2400,
          2500
        ]
      ],
      "enabled": true,
      "max_components": 2,
      "port_index": 0
    },
    {
      "bands_mhz": [
        [
          2400,
          2500
        ]
      ],
      "enabled": true,
      "max_components": 2,
      "port_index": 1
    },
    {
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        [
          2400,
          2500
        ]
      ],
      "enabled": true,
      "max_components": 2,
      "port_index": 2
    }
  ],
  "timeout_seconds": 120.0
}
```

This self-contained report is derived from an integrity-checked project snapshot. Exact recomputation also requires the original Touchstone and component model files identified by the project metadata.